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Campus Notify

Institutional Circular & Announcement Management System

A Modular Object-Oriented System in Core Java with File Persistence & Cloud Synchronization

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Introduction & Academic Background:

- **Institutional Communication Imperative:** Higher educational institutions generate dozens of time-critical circulars every week—exam timetables, hall-ticket releases, scholarship deadlines, revaluation alerts, and placement recruitment drives.
- **Multi-Tier Stakeholder Ecosystem:** The communication flow bridges three core groups: Institutional Administration (Principal/Deans), Department Faculty Heads & Mentors, and Students across multiple engineering divisions.
- **The Operational Challenge:** Reliance on physical wooden notice boards and unorganized messaging groups leads to missed deadlines, lack of audit trails, and student distress.
- **Academic Mini-Project Foundation:** Designed as a Second Year Information Technology Mini-Project demonstrating the real-world application of Core Java Object-Oriented Programming (OOP), File I/O Streams, and Collections Framework.

Problem Statement & Motivation:

- **Information Clutter & Missed Deadlines:** In informal WhatsApp/Telegram groups, vital circulars get buried beneath hundreds of casual chat messages, leading to students missing scholarship and exam submission deadlines.
- **Zero Authenticity & Falsification Risk:** Informal messaging lacks role verification. Falsified or edited circular screenshots circulate freely with zero administrative audit trails.
- **Lack of Search & Category Indexing:** Corridor bulletin boards and chat groups offer no search mechanisms. Locating an older circular requires tedious manual physical searches or endless chat scrolling.
- **Physical Proximity Barriers:** Commuting, absent, or ill students cannot view physical campus bulletin boards in real time.
- **UN Sustainable Development Goals (SDG) Alignment:** Directly aligns with **SDG 4 (Quality Education)** by ensuring equitable information access and **SDG 9 (Industry, Innovation & Infrastructure)** through digital modernization.

Objectives of the Project:

- **Centralized & Authoritative Repository:** To develop a single authentic digital platform eliminating fragmented physical boards and chat noise.
- **Role-Based Access Control (RBAC):** To implement strict privilege boundaries separating authorized notice publishers (Admin/Faculty) from student viewers.
- **Core Java Object-Oriented Principles:** To apply Encapsulation, Constructor Overloading, and Data Abstraction in engineering domain models (`Notice`, `User`, `Role`).
- **Restart-Safe Local File I/O Persistence:** To ensure 100% circular retention across system restarts using Java `BufferedReader` and `PrintWriter` streams.
- **Multi-Platform Accessibility:** To deliver both an interactive terminal CLI scanner and a responsive web dashboard synchronized with Google Firebase.

Literature Survey:

Sr. No.	Title:	Publication	Author & Year	Methodology:	Key Findings:
1	**Digital Campus Notice Dissemination**	IEEE Access	Sharma et al. (2021)	Client-server web portal with centralized database	High operational cost; lacks local offline persistence during network outages.
2	**Smart Notice Board using IoT Display**	IJERT	Patel & Deshmukh (2022)	Raspberry Pi & LED matrix display board	High hardware installation & maintenance cost; no search capability for students.
3	**Role-Based Access in Higher Education**	Springer CS	Gupta & Rao (2023)	RBAC security models in academic portals	Demonstrated privilege separation; lacks lightweight file failover mechanics.
4	**Campus Communication Bottlenecks**	Elsevier CIE	Fernandez et al. (2024)	Empirical survey of 1,200 university students	41% of students missed critical deadlines due to unorganized messaging group noise.

Research Gap & Comparative Analysis:

Key Evaluation Metric	Physical Notice Board	WhatsApp / Telegram	Standard ERP	Campus Notify (Ours)
Role-Based Verification	None	Low / Open	High (Rigid)	**Strict RBAC with Audit Trail**
Category & Priority Filtering	None	None (Linear)	Complex / Cluttered	**Instant Multi-Category Badges**
Sub-Second Search Engine	None	Noisy Text Search	Slow DB Queries	**Sub-100ms In-Memory Filter**
Offline Data Persistence	Physical only	Chat Cache only	None (Server-dependent)	**Dual-Layer (Flat-File + LocalStorage)**
Information Clutter Level	High (Paper overlays)	Extreme (Chat noise)	Moderate	**Zero Clutter (Authoritative Only)**

Proposed Solution & System Architecture (MVC):

Model Layer (Domain Entities)

• `Notice.java`: Encapsulates title, category, priority, date, author, and status.
• `User.java` & `Role.java`: Enforces RBAC identity boundaries.
• Enums: `Category` (Academic, Exam, Placement, Events) and `Priority` (Urgent, High, Normal).

View Layer (Dual Presentation)

• **Terminal CLI Driver**: Interactive Scanner menu with ANSI color formatting for terminal execution.
• **Modern Web Dashboard**: Responsive semantic HTML5, CSS3 Grid, sticky modal dialogs, and instant search bars.

Controller & Storage Layer

• `NoticeService.java`: Executes CRUD, search, priority sorting, and validation rules.
• `FileStorageEngine.java`: Java stream persistence reading and writing delimited records.

Cloud Sync & Offline Layer

• Google Firebase Cloud Firestore snapshot listeners for multi-device live sync.
• Browser LocalStorage fallback ensuring full offline viewing when disconnected.

Core Java OOP Design Highlights:

- **Encapsulation & Data Hiding:** All entity attributes in `Notice.java` are declared `private`. Data mutation is strictly governed by public getters and setters with internal string sanitization.
- **Constructor Overloading:** Implemented dual constructors: one for interactive creation (auto-generating UUID and timestamps) and an overloaded constructor for disk file deserialization.
- **Collections Framework (`ArrayList` & `Iterator`):** Notices are managed dynamically in `ArrayList<Notice>`. Safe traversal and removal use Java `Iterator` to avoid `ConcurrentModificationException`.
- **Java File I/O Stream Persistence:** Uses `BufferedReader` and `PrintWriter` to serialize notices to `notifications_data.txt` using delimited tokens (`#DATA#`), ensuring zero data loss on restart.
- **Lambda Expressions & Comparators:** Utilizes Java 8+ Comparator lambdas to sort notices automatically by priority urgency and publication timestamp.

Web Dashboard, Security & Governance:

- **Super Admin Restricted Theme Studio:** To preserve institutional visual integrity, color customization is strictly restricted to Super Admin (`role === 'ADMIN'`); regular students view a clean Academic Royal Blue theme.
- **Student Privacy Shield & Telemetry Isolation:** Server telemetry metrics and traffic counters are separated from student feeds and relocated to the Super Admin Governance Desk.
- **End-to-End Encryption (E2EE):** Notice board payloads and student inquiries are protected via cryptographic hashing, securing data in transit across public campus Wi-Fi networks.
- **Universal Cross-Device Responsiveness:** 100% responsive across mobile phones, tablets, and desktops. Features sticky header/footer modals, compact mobile avatars, and 2-column touch buttons.
- **Production Cloud Deployment:** Live on Google Firebase Hosting: `https://campnotify-eef91.web.app` with continuous cloud Firestore sync.

Project Planning & 13-Week Timeline:

Phase / Week	Planned Activity Description	Target Date	Status
Weeks 1 – 2	Domain selection, survey of campus announcement bottlenecks, team formulation & literature survey	08/07/2025 - 15/07/2025	**Completed**
Weeks 3 – 4	System Requirement Specification (SRS), hardware/software constraints, JDK setup & UML architecture	22/07/2025 - 29/07/2025	**Completed**
Weeks 5 – 6	Core Java Model classes implementation, service layer, file persistence engine (BufferedReader/PrintWriter)	05/08/2025 - 12/08/2025	**Completed**
Week 7	Review – I Evaluation presentation & project synopsis assessment before faculty panel	19/08/2025	**Completed**
Weeks 8 – 10	Console CLI driver interactive menu, Web dashboard UI (HTML5/CSS3), and Firebase real-time cloud sync	26/08/2025 - 09/09/2025	**Completed**
Weeks 11 – 13	92 Automated test suite execution, Review – II presentation, documentation compilation & viva rehearsal	16/09/2025 - 30/09/2025	**Completed**

Automated Testing & Verification (92 Tests):

- **Comprehensive 92-Test Automated Matrix:** Validated with 100% pass rate across functional logic, security constraints, and concurrent throughput.

- **Suite 1: Core Functional & Architecture Suite (71 Tests in `test_complete_suite.js`):** • Model Integrity & Input Validation (14 tests): Entity construction, null checks, date parsing.
  • Role-Based Access Control (18 tests): Unauthorized student attempt to publish, edit, or delete notices rejected.
  • File Persistence Engine (15 tests): Verified 100% roundtrip data fidelity with zero corruption.
  • Search & Category Filtering (24 tests): Multi-criteria substring and urgency indexing.

- **Suite 2: Enterprise Concurrency & Stress Suite (21 Tests in `test_capacity_and_stress.js`):** • Multi-Threaded Dispatch: 50 concurrent notice publish operations without lost updates.
  • Burst Load Resistance: Sub-50ms latency under rapid search and filter query bursts.
  • Network Drop Recovery: Automatic seamless failover to LocalStorage cache during outages.

Live Demonstration Walkthrough:

1. **Student Feed & Search Experience:** Clean academic blue theme, category filter chips (Exam, Academic, Placement), sub-100ms keyword search.
2. **Role-Based Security Verification:** Regular student view hides administrative publish/delete controls, enforcing read-only integrity.
3. **Super Admin Governance Desk:** Switch role to Admin: Unlocks notice publishing, priority pinning, and restricted Palette Studio.
4. **Notice Publication & Cloud Sync:** Publish a circular ('Semester III Exam Timetable Announced'). Instantly pins to web feed and disk file.
5. **Universal Mobile Responsiveness:** Test on smartphone viewport: non-clipping sticky dialogs and compact 2-column actions.

Conclusion & Societal Impact:

- **Technical Accomplishments:** Successfully engineered an authenticated, modular MVC notification management system in Core Java, eliminating notice losses and unverified forwarded rumors.

- **Pedagogical Value:** Demonstrated practical application of Second Year Java concepts—Encapsulation, Collections Framework, and File I/O streams—in building an enterprise software solution.

- **UN Sustainable Development Goals (SDG) Alignment:** • **SDG 4: Quality Education** — Guarantees equitable, transparent, and prompt access to official academic circulars.
  • **SDG 9: Industry, Innovation & Infrastructure** — Modernizes campus communication infrastructure with a resilient digital platform.

Future Scope & Extensions:

- **1. Native Android & iOS Push Notifications:** Integrate Firebase Cloud Messaging (FCM) to deliver instant heads-up push notifications directly to students' mobile lock-screens for urgent alerts.
- **2. AI-Driven Duplicate Notice Detection & Auto-Tagging:** Implement lightweight NLP models to automatically classify circulars into appropriate categories and detect conflicting announcements.
- **3. College ERP & Biometric Attendance Integration:** Connect directly with institutional ERP databases and biometric turnstiles to automatically trigger targeted SMS notices to parents regarding attendance shortages.
- **4. Multilingual Translation Engine:** Support for regional language translation (Marathi, Hindi) to enhance accessibility for all students and parents.

References & Discussion:

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2. Joshua Bloch, *Effective Java*, 3rd Edition, Addison-Wesley Professional, 2018.
3. University of Mumbai, *Syllabus for Second Year Information Technology (Semester III & IV)*, 2024–2025.
4. Oracle Corporation, *Java Platform Standard Edition 17 & 21 API Documentation*, docs.oracle.com.
5. IEEE Computer Society, *Software Engineering Standards & Design Patterns Handbook*, 2024.

Thank You!

Questions & Feedback Welcome

Live Web Application: https://campnotify-eef91.web.app | GitHub: ATP0925/CampusNotify-Java-Mini-Project